London Business School

Mergers, MBOs & Other Corporate Reorganizations

Constantine Yannelis

London
Business
School

Review of valuation methods

Overview

- Discounted cash-flow valuation methods
  - Free cash-flow
  - WACC method
  - APV method
  - Comparing APV and WACC
  - Equity cash-flow
- Valuation using Multiples
- Appendix: Computing Asset Betas and Equity Betas

Weighted Average Cost of Capital method

Two different FCF methods

Two different discounted cash-flow methods for total firm valuation with financing:

- Weighted Average Cost of Capital (WACC)
- Adjusted Present Value (APV)

Weighted Average Cost of Capital method

1. Determine the free cash-flow (FCF) of the investment

2. Compute the weighted average cost of capital (WACC)

$$r_{WACC} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D (1-\tau)$$

3. Compute the enterprise value (EV) by discounting the FCF of the investment (firm) with the WACC

$$V_{WACC} = EV = \sum_t \frac{FCF_t}{(1+r_{WACC})^t}$$

   – More debt usually means lower cost of capital (WACC) and therefore higher value

Weighted Average Cost of Capital method

- FCF = Cash-flow from operations + Cash-flow from investments
- $FCF = EBIT(1-\tau) + D\&A - \Delta NWC - Capx$

Revenue
– COGS
= Gross Profit
– Operating Expenses (e.g., SG&A)
= EBITDA
– D&A (often included in SG&A)
= EBIT/Operating Profit/Operating Income
– Taxes
= EBIAT/NOPAT
+ D&A
– Capx
– ΔNWC
= Free Cash-flow (FCF)

“Technical” notes:

- Non-cash expenses should be summed-up to the FCF
  - Typical example: depreciation should be “added back” because it is a non-cash expense
  - Similar: deferred taxes are a non-cash expense (this is tax that we owe, but do not have to pay presently)
- Non-cash (deferred) income should be subtracted from the FCF

- Taxes in FCF calculation: This is the amount of taxes paid if the business did not have debt (in the FCF calculation, we compute “Operating Taxes” off of EBIT)
  - Remember that the interest tax shield is considered in the discount rate (WACC), no double-counting

- Cash in net working capital (NWC)
  - Cash considered is *operating* cash: necessary to make payments related to the business
  - Non-operating (“excess”) cash (cash that could be used to pay back debt or pay a dividend): Not considered in valuation

Example: Using below accounting data, estimate the FCF for Disney in 2017 ($ mil.)

Walt Disney Co.

Key Financials

In millions of reported currency except per share data

Fiscal Year	**FY2012**	**FY2013**	**FY2014**	**FY2015**	**FY2016**	**FY2017**
Fiscal period length	12 months	12 months	12 months	12 months	12 months	12 months
Fiscal year end	Sep-29-2012	Sep-28-2013	Sep-27-2014	Oct-3-2015	Oct-1-2016	Sep-30-2017
Currency	USD	USD	USD	USD	USD	USD
Total Revenue	42,278	45,041	48,813	52,465	55,632	55,137
Growth Over Prior Year	*3.4%*	*6.5%*	*8.4%*	*7.5%*	*6.0%*	*(0.9%)*
EBITDA	10,971	11,642	13,828	15,578	16,987	16,770
Margin %	*25.9%*	*25.8%*	*28.3%*	*29.7%*	*30.5%*	*30.4%*
Depreciation & Amort., Total	1,784	1,957	2,064	2,132	2,320	2,586
EBIT	8,984	9,450	11,540	13,224	14,460	13,988
Margin %	*21.2%*	*21.0%*	*23.6%*	*25.2%*	*26.0%*	*25.4%*
Net Income	5,682	6,136	7,501	8,382	9,391	8,980
Margin %	*13.4%*	*13.6%*	*15.4%*	*16.0%*	*16.9%*	*16.3%*
Effective Tax Rate	33.3%	31.0%	34.6%	36.2%	34.2%	32.1%
+Total Current Assets	13,709	14,109	15,169	16,758	16,966	15,889
- Cash And Equivalents	3,387	3,931	3,421	4,269	4,610	4,017
= Operating Current Assets	10,322	10,178	11,748	12,489	12,356	11,872
+ Total Current Liabilities	12,813	11,704	13,292	16,334	16,842	19,595
- Short-term Borrowings	2,050	0	50	2,430	1,521	2,772
- Curr. Port. of LT Debt	1,564	1,606	2,114	2,133	2,166	3,400
= Operating Current Liabilities	9,199	10,098	11,128	11,771	13,155	13,423
Net (operating) Working Capital	1,123	80	620	718	-799	-1,551
Change in Net Working Capital	-416	-1,043	540	98	-1,517	-752
Capital Expenditure	-3,784	-2,796	-3,311	-4,265	-4,773	-3,623
Cash Acquisitions	-488	-2,443	-402	0	0	-417
Divestitures	0	0	0	0	0	0

Example: Disney's FCF

Data	FY2017				
Sales =	55,137	$M			
EBITDA =	16,770	$M			
EBIT =	13,988	$M			
Tax rate =	32.1%				
Operating taxes =	4,486	$M			
Capital spending =	4,040	$M			
Depreciation =	2,586	$M			
Increase in Non-cash WC =	-752	$M			
FCF = EBIT*(1-tax rate)	9,502	=	13,988	*	67.9%
− Net Capex	1,454	=	4,040	-	2,586
− Change in WC	-752				
Free Cashflow =	8,800				

Weighted Average Cost of Capital method

- Cost of debt, aka required return on debt (r_D)
  - Can be estimated using return on a traded, long-term obligation (bond)
    - YTM of comparable bond (though YTM > true r_D if probability of default high)
  - Can also use spread over treasuries for bond indices with given debt ratings
  - If β_D known, can calculate r_D using CAPM

- Cost of equity, aka required return on equity (r_E)
  - Can use CAPM: $r_E = r_F + \beta_E * MRP$
    - Directly use firm's CAPM β_E (if capital structure is stable, and firm is listed)
    - Alternatively, start with industry asset Beta β_A, re-lever to match target capital structure (if capital structure expected to change in future and/or firm is not listed)
  - More on this later (see APV method discussion)

Weighted Average Cost of Capital method

WACC capital structure weights ($\frac{E}{E+D}$ and $\frac{D}{E+D}$): how to compute them

- Typically, WACC is used to compute the value of equity (or the enterprise value *E* + *D*)…
  … but you need to know *E* if you want to compute *D/E* !
- Thus, the capital structure weights reflect the (constant) target leverage ratio of the firm, so we do not need to know *E* or *E+D* to calculate the weights
- *Technical note:* conceptually, the "D" in the capital structure weights reflects *net debt*, that is, debt less any *excess* cash

Adjusted Present Value method

The Adjusted Present Value method

- An alternative approach to WACC method is to use Adjusted Present Value (APV) method to calculate the enterprise value (EV)

- Three simple steps involved in computing the APV:
  - **Step 1**: Value of the firm as if all-equity financed: use the after-tax cash flows (FCF) and discount them at the "asset" cost of capital r_A (aka unlevered cost of capital)
    - For an all-equity firm (only), this cost of capital equals the cost of equity r_E
  - **Step 2**: Add the present value of the tax shield (*TS*) generated by the firm over its life
  - **Step 3**: Can also account for other financing effects (if any), such as the present values of costs of financial distress, loan subsidies, and various financial fees

$$V_{APV} = EV = V_{\text{all equity}} + PV(\text{Financing effects})$$

The Adjusted Present Value method

- To be clear (what is meant by "PV(Financing effects)" on previous slide):

$$V_{APV} = EV = V_{all\ equity} + PV(TS) - PV(CFD) + PV(other\ effects)$$

- APV method is also known as a "valuation by components"

- In the simpler form, we consider only the tax shield as a financing effect:

$$EV = V_{all\ equity} + PV(TS)$$

The Adjusted Present Value method

- The APV method uses the “value additivity principle” to evaluate the contribution of both cash flows and increased debt tax shields.
  - It can easily be adapted to include other financing side effects.

- We do this to separate the value of running the business from the value created by financing
  - More debt means tax savings from debt because interest is tax-deductible
  - (But too much debt will increase the costs of financial distress in expectation)

- This separation of the value into different components allows us to identify the sources of value and to discount different risks appropriately

APV: Inputs and steps required

- Forecast the cash-flows from running the business as an all-equity firm for 5 or 10 years (until you get to a stable period, the “steady state”)
- Discount these cash-flows at the appropriate rate for the risk of running the business
- Account for the terminal value and discount it

Let’s analyze in detail each of the steps involved and the inputs required…

Which discount rate for the FCF?

- You need the rate that would be appropriate to discount the firm's cash-flows if the firm were 100% equity financed.
- This rate is the expected return on equity if the firm were 100% equity financed.
- To get it, you need to:
  - Find comparables, that is, publicly traded firms in same business.
  - Estimate their expected return on equity if they were 100% equity-financed, by "unlevering" the equity Betas
  - See the Appendix of this document for details

Which discount rate?

- Use the comps' average or median asset Beta (β_A) as an estimate of the project's (or firm's, if you are valuing a whole firm) $\boldsymbol{\beta_A}$

- Use the estimated β_A to calculate the all-equity (asset) cost of capital r_A

$$r_A = r_F + \beta_A * \text{Market Risk Premium}$$

- Use r_A to discount the firm's FCF

- What's the risk-free rate?
  - Short-term or long-term rate r_F? Historical average or forward looking r_F?
  - Ideally forward-looking: yield on long-term risk-free government bonds

Market Risk Premium

- The CAPM's Market Risk Premium (MRP) is unobservable
- Estimate of the Market Risk Premium (MRP): risk premium in equity markets

- Historical Equity Risk Premium
  - There are many estimates of risk premium of market over government bonds; Historically 6% (per annum)
- Implied Equity Risk Premium
  - Fama & French (2001): Risk premium implied by fundamentals and stock prices over the 1872-2000 period was between 3% and 5%
- Equity Risk Premium – Survey estimates
  - Graham and Harvey (2001) survey of CFOs: 10-year risk premium ranges between 4% and 5%.

Detour on the discount rate

- Best check on the appropriateness of the discount rate:
  - Sensitivity Analysis: what range in EV do you get if you vary the discount rate?
  - This, in fact, applies to any of the estimated inputs of a model

- Getting “fancy” with the discount rate is typically a low payoff activity.
  - Using “term structure” (different risk-free rates for each period) normally adds little.
  - Using Fama-French (three-factor or four-factor) model instead of CAPM may not be easy and is rarely done in a corporate valuation context

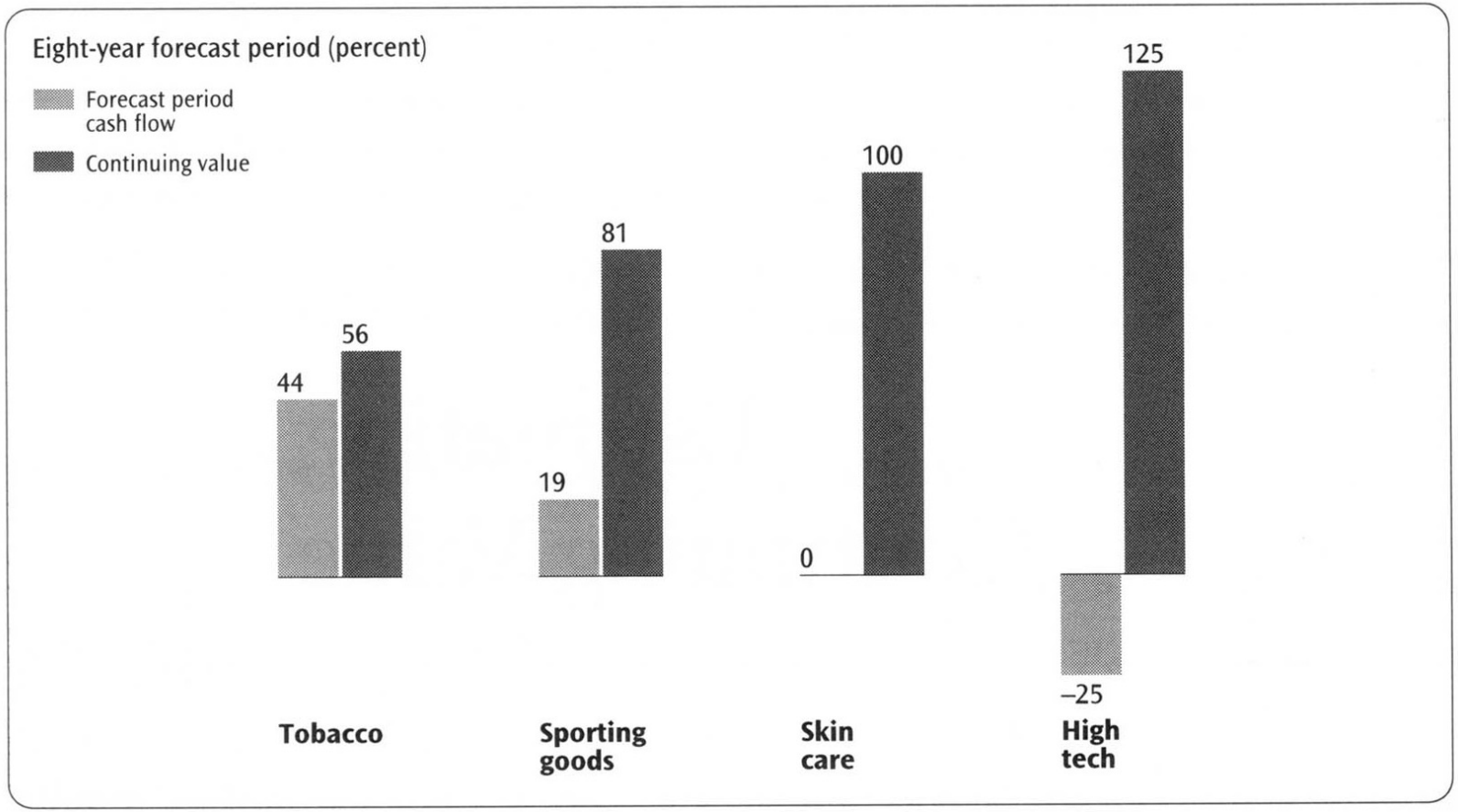
Terminal values

There are generally three ways to proceed to compute **terminal value (TV)**

- If project is liquidated or sold at the end of the horizon, we calculate its *salvage value* → "Liquidation method"
  - Should account for taxes on capital gains or tax shields on capital losses
- Use the latest detailed cash-flow forecast and assume a steady-state growth rate for the business to calculate the *continuation value* → "Perpetuity method"
- Assume that you can sell the firm using a simple rule involving earnings multiples → "Exit Multiples method"
  - More on Multiples later

Terminal values can be very large fraction of total firm value

Exhibit 12.1 Continuing Value as a Percentage of Total Value



Terminal values: Perpetuity calculation

- For the Perpetuity Method, we assume the firm is in steady state at time *T*.
  - *T* is the end of the detailed forecast horizon (typically 5 – 10 years)
  - Note: Assumption of steady-state growth may not be appropriate with high-growing companies, cyclical industries

- Use the last FCF forecast from year *T*, grow it by one period (recall PV formula), and divide it by r_A - *g.*

$$TV_T = \frac{FCF_T(1+g)}{r_A - g}$$

Terminal values: Perpetuity calculation

- To get *PV(TV)*, need to discount back: $PV(TV) = TV_T / (1 + r_A)^T$
- Make sure FCF_T used to get *TV* reflects steady-state growth *g*:
  - Adjust capital expenditures, depreciation, and changes in net working capital to reflect steady-state growth
  - Ask yourself: are depreciation, NWC and other flows you have in the cash-flow estimate consistent with the *g* that you choose?
- Steady-state growth rate *g*, typical choices considered:
  - Economy-wide (GDP) nominal growth rate
  - Inflation rate (assumes zero real growth)

Terminal Values: Multiples of earnings

- If you use a multiple of earnings to calculate the sale price, then you need to multiply earnings (not free cash-flows) with the multiple to get the *TV*
  - For example, you may estimate that you sell the business at a 10x EBITDA multiple in year *T*

- Where do you get the multiple from?
  - Pick the multiple using comparable firms or transactions
  - Be sure to perform sensitivity analysis
  - More on Valuation with multiples later

Step 2 in APV: Add *PV(Tax Shield of Debt)*

- Need to account for the debt tax shields, including any associated with the terminal value and discount these at the appropriate interest rate.

- Interest payments are tax deductible. Reduce tax payments by:

$$\tau * interest\ expense$$

  – These deductions depend on the interest payments and the corporate tax rate (τ)

Important caveats

- We only consider corporate taxation
  - We could also consider our investors' tax rates, but this is rarely done
- Debt Tax Shield is only valuable when taxes are paid (firm is profitable)
  - Otherwise, must keep track of NOLs and carrybacks (this can be complicated)

Discounting the tax shield

- The choice of the discount rate depends on the risk of the interest tax shield.

- In general, the tax shield will have its “own risk”
  - That is, the tax shield has its own Beta (β)
  - The risk will depend on the probability that interest payments will be made (default risk)
  - The risk depends also on possible changes in the amount of debt over time
  - It will also depend on the probability that the government will change its tax policy

Discounting the tax shield

- Precisely estimating the risk of the tax shield could be very difficult (and is often not worthwhile)

- Typically, we choose among two assumptions:
  - The risk of the tax shield is equal to the *risk of the debt* (β_D)
  - The risk of the tax shield is equal to the *risk of the assets* (β_A)

Discounting the tax shield

- If debt is predetermined or has a low level, the risk of tax shields is equal to risk of repayments to debtholders; so r_D is correct discount rate

- For example, when there is a permanent debt D, the present value of the interest tax shield is equal to τD
  - This is obtained by discounting a fixed, constant annual perpetuity (the annual interest tax shield) with r_D

Discounting the tax shield

- If the level of debt varies with firm value (for example, the firm has a policy to keep its leverage ratio constant)
- Or: the firm has a very high level of debt (e.g., LBO)

- Then: riskiness of tax shields is similar to that of operating assets and tax shields should be discounted at r_A

Discounting the tax shield

- Whatever assumption you make about the risk of the tax shield, you must take care to be **consistent** throughout the valuation.

- In particular, remember that the assumption you make about the risk of the tax shield also affects the equation you will use to **lever** and **unlever** the Beta.
  - See also: Appendix

SCENARIO 1 – Using r_A to discount debt tax shield

- Debt in the future is not fixed.
  - Debt and interest expense are "tied" to FCF

- Reasonable when:
  - LBO-type debt levels (high debt; risk of the debt is close to the risk of the assets)
  - Firm has constant leverage *ratio*

- In those cases, ability to use tax shields has more systematic risk than the ability to pay debt.

$$TS = \text{Tax Rate} * \left[\frac{Int_1}{(1+r_A)} + \frac{Int_2}{(1+r_A)^2} + \ldots + \frac{Int_T}{(1+r_A)^T} \right]$$

  - Int_t (expected interest payment in year t) determined by debt repayment schedule and interest rates

Two scenarios for discounting tax shield: r_A or r_D

SCENARIO 2 – Using r_D to discount debt tax shield

- You expect a predictable, stable, and low level of debt

- Ability to use tax shield has same systematic risk as ability to pay debt
  - Best suited in lower-leverage situations

$$TS = \text{Tax Rate} * \left[\frac{Int_1}{(1+r_D)} + \frac{Int_2}{(1+r_D)^2} + \ldots + \frac{Int_T}{(1+r_D)^T} \right]$$

- For permanent (fixed *amount*) debt with the same risk as interest:

$$TS = \text{Permanent debt} * \text{Tax rate} = \tau * D$$

Summary of tax shield discount rates

- **Remark 1:** If debt is predetermined (D is constant) and has a low level (low absolute amount), risk of tax shield is similar to risk of payments to debt holders; so r_D is the correct discount rate

- **Remark 2:** If high level of debt or if level of debt varies with firm value (leverage ratio D/V is constant), then the risk of the tax shield is similar to that of operating assets and tax shields should be discounted at r_A

- **Remark 3:** Some firms have low leverage ratios or maintain a target leverage ratio while maintaining an investment grade rating on their debt. For these firms, it can be argued that the correct discount rate is probably closer to r_D

Tax shield and the Betas

- Whatever assumption you make about the risk of the tax shield, you must be **consistent** throughout the valuation
  - Remember that the assumption you make about the risk of the tax shield also affects the equation you will use to lever and unlever the Beta (see Appendix)

- If the risk of the tax shield is equal to the risk of the *debt* (*D* is constant), then the unleveraging equation is:

$$\beta_A = \beta_E \frac{E}{D(1-\tau)+E} + \beta_D \frac{D(1-\tau)}{D(1-\tau)+E}$$

- If instead the risk of the tax shield is equal to the risk of the *assets* (e.g., *D/V* is constant), then the unleveraging equation is:

$$\beta_A = \beta_E \frac{E}{D+E} + \beta_D \frac{D}{D+E}$$

What about the tax shield associated with the TV?

- You must add any debt tax shields to NPV that accrue beyond the terminal date

- Typically, two different possibilities are considered:
  - Debt level (*D*) stabilizes at a permanent (and low) level, D_T, at the end of the "detailed forecast horizon" (*T*)

$$TV(TS) = \frac{\tau * r_D * D_T}{r_D} = \tau * D_T$$

  - Leverage *ratio* (*D/V*) stabilizes at some level at *T*:

$$TV(TS) = \frac{\tau * r_D * D_T}{r_A - g}$$

Extending the APV method

- The APV method is very flexible.

- Other potential financing effects one could include (aside from the standard interest tax shield) are loan guarantees, subsidized loans, costs of issuing securities, etc.

- One could also account for costs of financial distress, agency costs, etc. How?
  - Write scenarios and include costs of distress in bad scenarios
  - In practice, PV(costs of distress) and PV(agency costs) are typically ignored because they are very difficult to quantify

- If you want to include any of these costs, calculate the present value and add or subtract it from the all-equity NPV.

APV vs WACC method

Comparing methods: APV

- **Pros:**
  - Implicit assumptions are very clear
  - Works for complicated debt schedules (difficult with WACC method)
  - Easy to use if the level of future debt is known
  - Transparent method: Puts the spotlight on what is creating value; easier to track where value comes from
  - Assists in the decision of how to structure financing for projects
  - More flexible:
    - Just add other financing effects as separate terms
    - Allows to use different discount rates for different components

- **Cons:**
  - Requires explicitly deriving schedule of future debt levels and interest payments
  - Increasingly used (PE), but still less than WACC

Comparing Methods: WACC

Pros:

- Most widely used
- The inputs are usually relatively easy to get
- If you have a precise target debt-to-value (constant leverage ratio) policy, then it is easy and accurate
  - This assumption may be overly restrictive in practice
- Requires fewer, often simpler computations

Comparing Methods: WACC

Cons:

- "Lumps together" the effects of assets (operating effects) and liabilities (financing effects)
  - Errors/approximations in effect of liabilities will contaminate the whole valuation (e.g., modelling of wrong leverage policy)
  - Does not directly reveal where the value is coming from
- Not very flexible; cannot easily allow for changes in:
  - Other effects of financing (e.g., costs of distress, issuance costs)
  - Non-constant debt ratios
  - Changes in tax rates

Practical implications

- In principle, we can always use either WACC or APV
- As long as you follow all the steps, and you are careful in making the same assumptions you should obtain (approximately) the same values

- In practice, however, one of the two will be much simpler to use depending on the situation
  - For example: for complex, changing or highly leveraged capital structure (e.g., LBOs), APV is typically more suited

- To illustrate these points, let's consider two cases:
  - Debt is rebalanced regularly (annually) to maintain a *fixed leverage ratio*
  - *Debt levels (not ratios) are predetermined*, implying changes in the leverage ratio over time

Practical implications

- If firm has a target (constant) debt-to-asset ratio:
  - Computing WACC is much easier
  - APV becomes more complex since *D* will change every period (year)
  - **Bottom Line**: WACC preferred

- If debt is predetermined (firm knows the evolution of *D;* could keep it constant):
  - APV is easy to compute by discounting future expected interest tax shields
  - WACC instead is difficult to apply, because if the debt is not rebalanced and D/V changes over time, then so does WACC
    - You can always forecast D/V values, re-compute a different WACC for each year and discount; but: lots of opportunities for errors
  - **Bottom Line:** APV preferred

Flow-to-equity method

Equity cash-flow

- The equity cash-flow (ECF) measures the cash-flow available to shareholders after payments to debt holders are deducted from operating cash-flows

- Payment to debt holders are sometimes called “Debt Cash Flows” (*DebtCF*), and they include interest and principal payments

$$ECF = CCF - DebtCF$$

- As *DebtCF* are paid out of operating cash flows before equity cash-flows, debt cash-flows are less risky than equity cash-flows

- ECF are riskier than cash-flow measures that combine DCF and ECF, which implies higher discount rates

Equity cash-flow

- ECF are calculated by subtracting taxes, interest and debt repayments from operating cash flows and adding debt increases

  ECF = Net Income + Depreciation – Capx – Change in NWC + Change in Debt

- The discount rate used for the ECF is the *expected return on equity* (r_E) and the value we obtain is the *equity value* of the firm

- When the leverage ratio is not constant over time, this approach requires that a new equity cost of capital be calculated each year of the forecast (same problem as in WACC method)
  - Complicated, and lots of opportunities for errors

Overall comparing methods

- In general, we typically prefer both APV and WACC to ECF
- APV is preferred to ECF for many of the same reasons that APV is superior to WACC
  - Use of one constant discount rate for ECF requires that the firm has a constant debt to total capital (leverage) ratio
    - If the debt to total capital ratio varies over time, we need to calculate a different discount rate for the equity for each year
  - It is easier to make mistakes using the ECF method
    - In particular, it is critical to include increases in debt in the cash-flows, particularly in the terminal value calculation

Valuation using multiples

Valuation using multiples

- Main idea: value the target asset (firm) by using information on the pricing of other similar assets (or transactions)
  - Also called “relative” or “comparables” valuation

$$\text{Value}_{Firm} = \left[\frac{\text{Value measure}}{\text{Key characteristic}}\right]_{Industry\ average} * \text{Key characteristic}_{Firm}$$

- Numerator of multiple is a measure of total value of a firm
  - e.g., price, enterprise value

- Denominator of multiple is the characteristic that is important for that firm, often industry-specific: EBITDA or more ”funky” characteristics like miles flown (airlines)

Valuation by multiples: Implicit assumptions

1. Comparable companies assumed to have expected cash-flows *growing at the same rate* and have the *same level of risk* as the company being valued

2. Value of the company assumed to vary in direct proportion with changes in the performance measure; i.e., if expected EBITDA increases by 8%, expected firm value also increases by 8%

3. Is the kind of transaction involving the comparable similar to ours?
   - Price may be different for buying one vs. many shares
   - It may also be different in liquid vs. illiquid markets
   - Was the price of the transaction influenced by a special context?

4. Is the “multiplied” variable (e.g., EBITDA) a good proxy for future performance?
   - Can it be too high/too low today just by accident?
   - Forward looking projections (e.g., analyst consensus forecasts) more relevant than historical figures

Types of multiples

- Cash-flow-based *Value* multiples:
  - EV/Earnings, EV/EBITDA, EV/FCF
- Cash-flow-based *Price* multiples:
  - Price/Earnings (P/E), Price/EBITDA, Price/FCF
- Asset-based *Value* multiples:
  - EV/BV of assets, MV of equity/BV of equity
- Industry-specific *Value* multiples:
  - EV/Hospital beds, EV/Number of customers

Motivation for multiples

- Firms in the same line of business typically have similar multiples

- If the firm's actual FCF is a growing perpetuity:

$$EV = FCF/(WACC-g) \Rightarrow EV/FCF = 1/(WACC-g)$$

- Comparables will have a similar EV/FCF provided they:
  - Have the same WACC (requires similar leverage ratio)
  - And are growing at a similar rate

Procedure (“trading” multiples)

- Step 1: Identify firms in same business as the firm you wish to value. Selection should be representative; be mindful of ”outliers”
  i. Search in large databases (e.g., Bloomberg)
  ii. Progressively exclude companies based on criteria such as:
     - finer industry
     - size
     - geography
     - diversification
  iii. Apply more subjective criteria such as:
     - market position
     - life-cycle stage
     - differences in customer bases
     - etc.

Procedure (“trading” multiples)

- Step 2: Calculate the multiple(s) for comparable firms
- Step 3: Calculate average multiple for the set of comparable firms
  - This gives an estimate of the multiple we can use in valuing our firm
  - If you are worried about outliers: use median
  - If you trust some estimates more than others, consider value-weighting different multiples when calculating the average
- Step 4: Apply the multiple on the firm you wish to value
  - Often different multiples provide somewhat different answers
  - Use judgment on what is economically reasonable

Implicit assumption: Firms in the same industry have similar multiples

Remarks

- When choosing comparable firms, you face a bias-variance trade-off:
  - Too many firms leads to including firms in the multiple calculation that are not truly comparable
  - Too few firms means that your average will reflect idiosyncrasies of the comparable firms

- For firms with no earnings or limited asset base (e.g., hi-tech):
  - Price-to-patents multiples
  - Price-to-subscribers multiples
  - Also: website clicks, number of engineers (PhD employees), etc.

- Multiples based on equity value (or stock price, e.g., P/E) as opposed to total firm value ignore effect of leverage on the cost of equity (or assume the firms have similar leverage)
  - Beware if comparables have very different leverage

Remarks

- In the case of transaction multiples (e.g., when calculating the sales price of company):
  - Use recent transactions that are comparable in type (merger, LBO etc.), industry, size
  - Account for deal specificities like existence of a control premium or liquidity discount
  - Be aware, everyone might be over/underpaying (is the market “right”?)

Pros and cons of multiples

Pros:

- Incorporates a lot of information from other valuations in a simple way
- Embodies market consensus about discount rate and growth rate
- Use market information
- Can provide discipline in valuation process by ensuring that your (DCF) valuation is in line with other valuations
- Sometimes, what you care about is what the market will pay, not the fundamental value

Pros and cons of multiples

Cons:

- Difficult to find *true* comparables
  - Implicitly assumes all comparables are alike in growth rates, cost of capital, and business composition
- Hard to incorporate firm-specific information
  - Problematic if operating changes will be implemented in the target firm
- Relies on accounting measures being comparable
- Differences in accounting practices can affect earnings and equity-based multiples
- If market is overpaying, you will too

Troubles with multiples

- Remember: The value of an investment is the present value of the expected future cash flows. *That's it.*

- Construction company executive: *"For property development we use multiples, not present values. We buy a property, develop it, and sell it 3 months later. We are interested in the price we can get when we sell it. Not the actual value of the property."*

- Investment bank executive: *"We never use present values, only multiples. (…) Our business is to assist in selling of companies, clients pay us for our large network of contacts with potential buyers"*

Bottom line on multiples

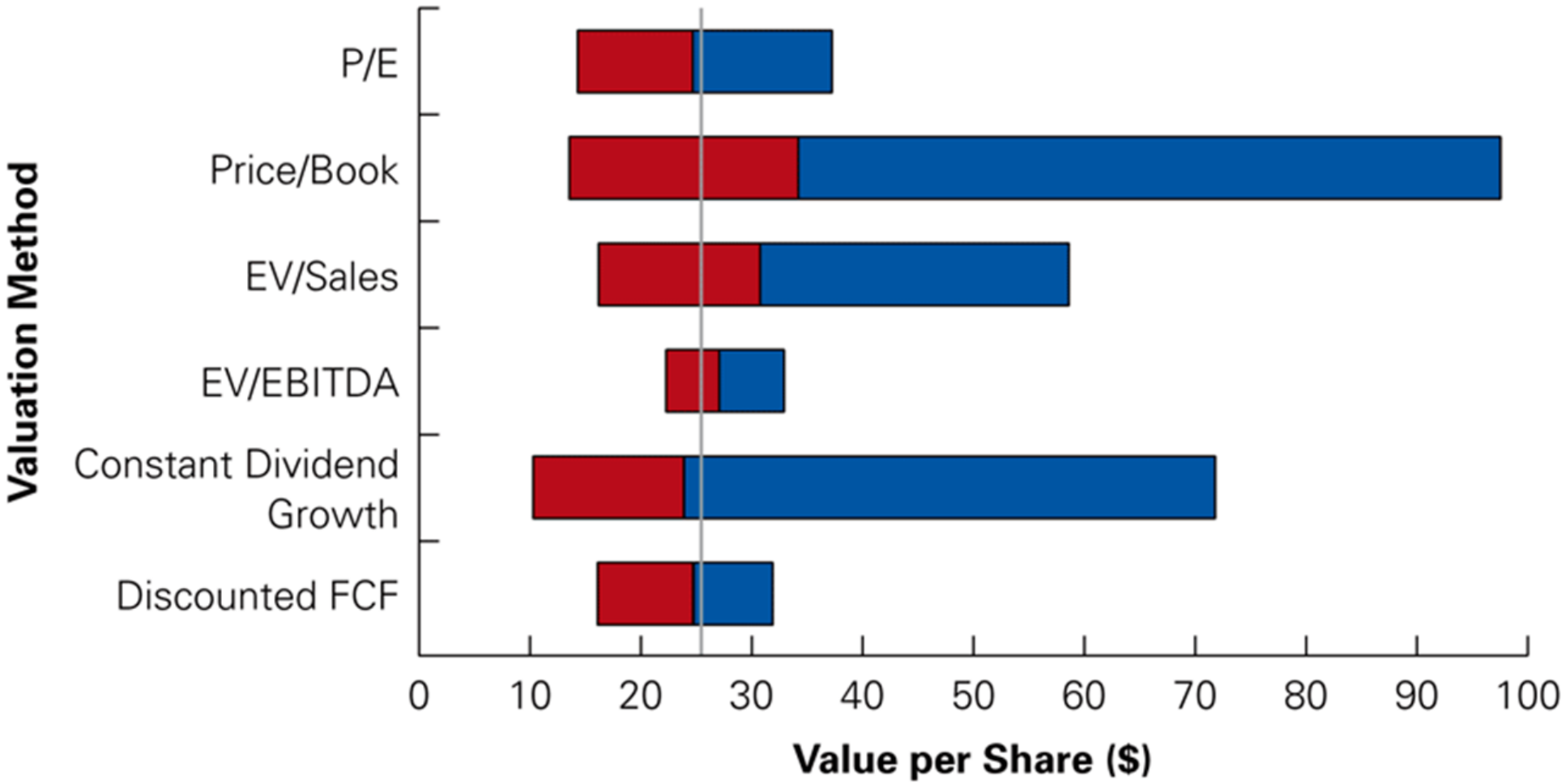
- Multiples **complement DCF** methodology:
  - Useful check on valuation; market-based perspective
  - Useful when you do not have cash-flow projections (“back-of-envelope” valuation)

- **EBITDA or cash-flow multiples are preferable** to (net) earnings multiples
  - More consistent treatment of leverage
    - Companies with different leverage will have different P/E multiples, even tough same business risk and growth
  - Lower ability to manipulate
    - Easier to manipulate net earnings through accounting than EBITDA or cash flow

Bottom line on multiples

- Because of the many limitations, never rely on just a single multiple or on valuation based only on multiples
- Best to use multiples (only) as a **check for the valuation based on discounted cash-flows**
  - After you have done a thorough valuation, you can compare the multiples implied in your valuation, such as P/E and market-to-book, to representative multiples of similar firms.
  - If your predicted multiples are very different, then you must convince yourself (and your clients) that your model is reasonable.
- Because of accounting differences, be careful when using multiples to compare firms across industries, and especially, across countries
  - As a rule of thumb when choosing the basis for multiples: The higher up the basis is in the income statement (e.g., sales) the less it is subject to accounting idiosyncrasies

Valuation “Football field”



Appendix:
Unlevering and relevering Betas

Unlevering and relevering Betas

- Unlever each comparable firm's β_E to estimate its Asset Beta (β_E)
- Consider the following simplified market value-based balance sheet

Operating assets (A)	Debt
PV(ITS)	Equity

$$A + PV(ITS) = E + D$$

- We can use the balance sheet analogy also for the risks (Betas) of the firm

Unlevering and relevering Betas

β_A	β_D
$\beta_{PV(ITS)}$	β_E

Beta of the firm is equal to the weighted average of the Betas:

$$\frac{A}{D+E}\beta_A + \frac{PV(ITS)}{D+E}\beta_{PV(ITS)} = \frac{E}{D+E}\beta_E + \frac{D}{D+E}\beta_D$$

Unlevering and relevering Betas

$$\frac{A}{D+E}\beta_A + \frac{PV(ITS)}{D+E}\beta_{PV(ITS)} = \frac{E}{D+E}\beta_E + \frac{D}{D+E}\beta_D$$

To get a useful expression for the asset risk of a firm (β_A), two assumptions are common

1. $\beta_{PV(ITS)} = \beta_A$
- Risk of tax shield cashflows is similar to operating asset cashflow risk

2. $\beta_{PV(ITS)} = \beta_D$
- Risk of tax shield cashflows is similar to debt payment risk

Unlevering and relevering Betas

If the **risk of tax shields is similar to operating asset risk**:

- Then: assume $\beta_{PV(ITS)} = \beta_A$
- Then we can express asset Beta as: $\beta_A = \beta_E \frac{E}{D+E} + \beta_D \frac{D}{D+E}$
- Can express levered equity Beta as:

$$\beta_E = \frac{\left(\beta_A - \beta_D \frac{D}{D+E}\right)}{\frac{E}{D+E}} = \beta_A + \frac{D}{E}(\beta_A - \beta_D)$$

- *This case is appropriate if a firm has a target (constant) leverage ratio (or "high" leverage)*

Unlevering and relevering Betas

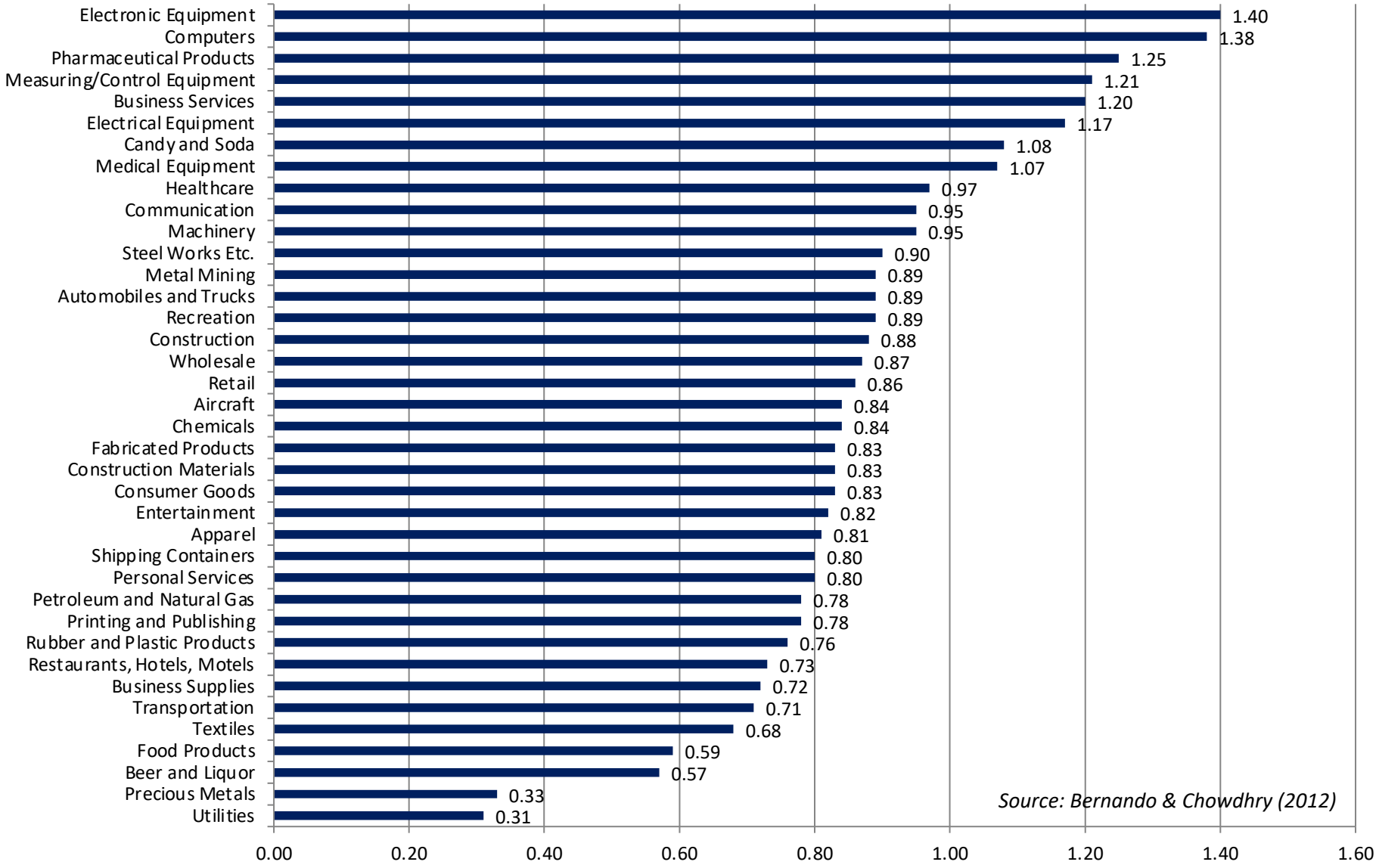
If the **risk of tax shields is similar to debt risk**:

- Then: assume $\beta_{PV(ITS)} = \beta_D$
- Then we can express asset Beta as:

$$\beta_A = \beta_E \frac{E}{D(1-\tau)+E} + \beta_D \frac{D(1-\tau)}{D(1-\tau)+E}$$

- If asset Beta is known, we can express levered equity Beta as:

$$\beta_E = \beta_A + (1-\tau)(\beta_A - \beta_D)\frac{D}{E}$$

- Appropriate if fixed (permanent) leverage (that is not “too high”)

Asset Betas across industries